

Project Initialization and Planning Phase

Date	2 July 2026
Team ID	xxxxxxx
Project Name	Rising waters
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Floods are among the most frequent and destructive natural disasters, causing loss of life, damage to infrastructure, and economic setbacks. Traditional flood prediction methods often rely on manual monitoring and may not provide timely warnings. The objective of this project is to develop a **Machine Learning-based Flood Prediction System** that analyzes environmental factors such as rainfall, river level, humidity, temperature, soil moisture, drainage, and elevation to predict flood risk accurately. The system aims to provide early warnings and support disaster management authorities in making informed decisions.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A resident living in a flood-prone area	Know whether my area is at risk of flooding..	I don't receive accurate flood warnings in advance.	Flood conditions depend on changing weather and environmental factors.	Anxious and unprepared for emergencies.
PS-2	A disaster management officer	Predict flood risk and plan rescue operations.	It is difficult to analyze large amounts of weather and river data quickly.	. Manual monitoring is slow and prone to errors.	Stressed about making timely decisions.
PS-3	A government authority	Reduce flood-related damage and improve disaster preparedness.	I don't have reliable, real-time flood risk predictions. .	Floods are influenced by multiple environmental factors that change rapidly.	Concerned about protecting lives and infrastructure.

Solution Statement

Developed an intelligent **Flood Prediction System** that uses **Machine Learning** to analyze environmental parameters such as **rainfall, river level, humidity, temperature, soil moisture, drainage, and elevation** to accurately predict flood risk. The system provides early warnings and supports disaster management authorities and citizens in making informed decisions, helping reduce loss of life, property damage, and improving disaster preparedness.